MANIPAL ACADEMY of HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

**MANIPAL SCHOOL OF INFORMATION SCIENCES**

**(A Constituent unit of MAHE, Manipal)**

# Linux OS and Scripting – 3_Theory

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|---|---|---|
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**04/09/2026**

**Exercise 1 – Identify Your Linux Filesystem**

**Explore the storage configuration of your Linux system.**

**Determine:**

**1. The storage devices available.**

You can use lsblk to see the storage devices attached to a Linux system. Its output lists the disks and other block devices connected to the machine.

**2. The partitions available on each device.**

lsblk also shows the partitions on each device, letting you identify how the main disk is split up.

**3. The filesystem type used by each partition.**

Use lsblk -f to find each partition's filesystem type. Common examples are ext4 for Linux filesystems and swap for swap space..

**4. The device/partition on which `/` is located.**

df -T shows which device the root filesystem / is mounted on. The partition shown next to / is the one holding the root filesystem.

**5. The filesystem type used for `/`.**

df -T also reveals the filesystem type for /. On a standard Kali Linux install, this is usually ext4.

**6. The total, used, and available storage space.**

df -h shows total filesystem size, used space, available space, and usage percentage. These values should be taken from the user's own system output.

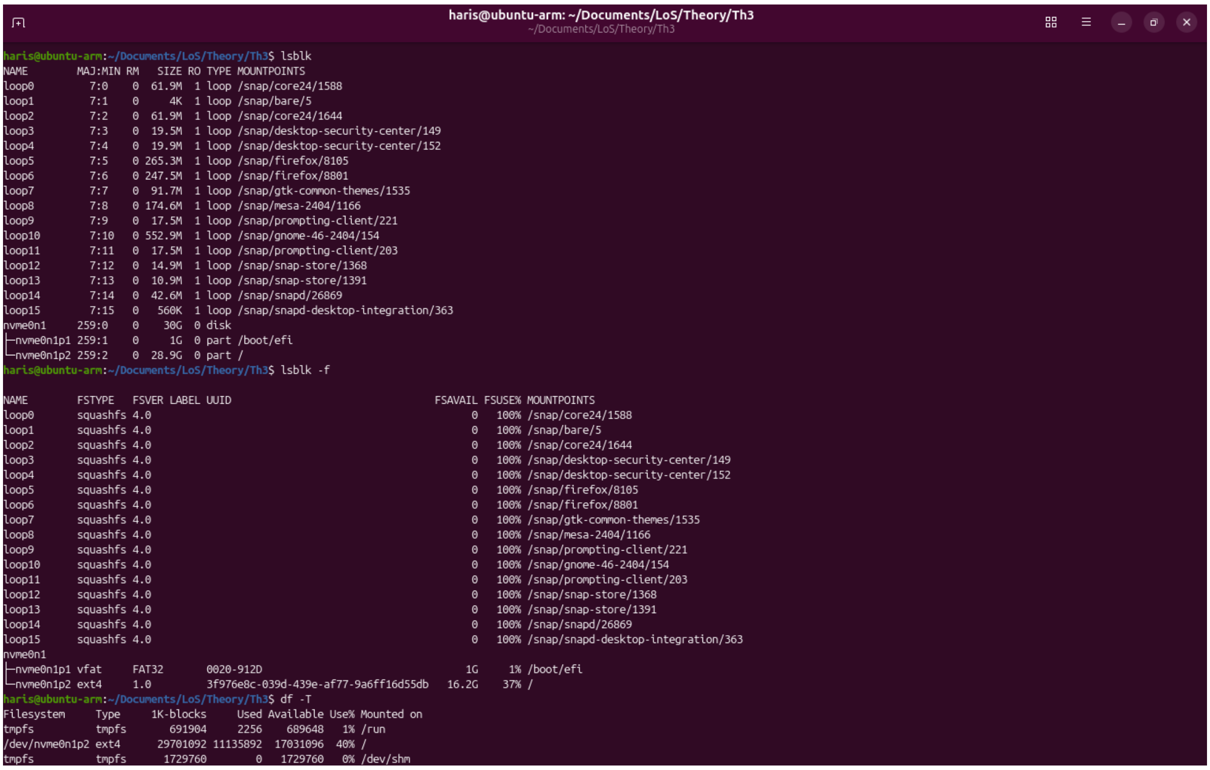
****Suggested commands:****

**lsblk**

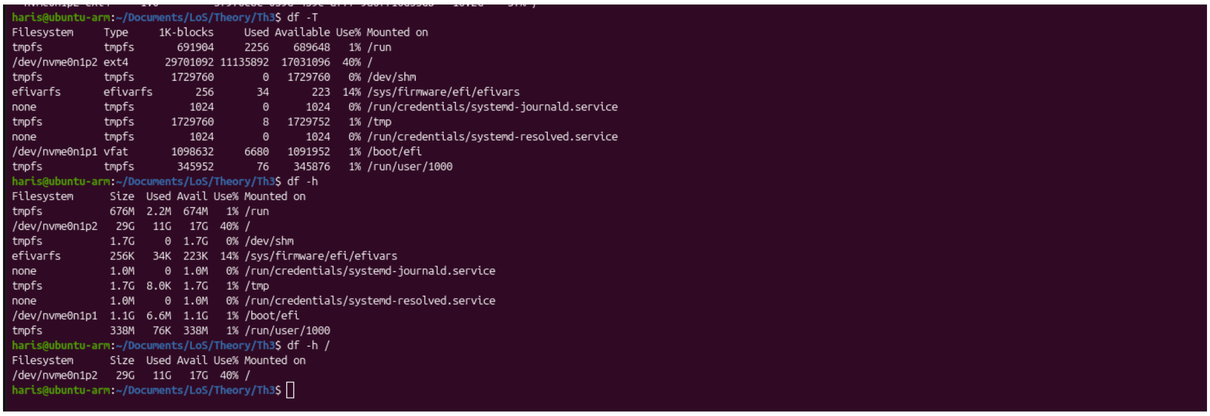
**lsblk -f**

**df -T df**

**-h**



```
haris@ubuntu-arm: ~/Documents/LoS/Theory/Th3
~/Documents/LoS/Theory/Th3
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0         7:0    0  61.9M  1 loop /snap/core24/1588
loop1         7:1    0     4K  1 loop /snap/bare/5
loop2         7:2    0  61.9M  1 loop /snap/core24/1644
loop3         7:3    0  19.5M  1 loop /snap/desktop-security-center/149
loop4         7:4    0  19.9M  1 loop /snap/desktop-security-center/152
loop5         7:5    0 265.3M  1 loop /snap/firefox/8105
loop6         7:6    0 247.5M  1 loop /snap/firefox/8801
loop7         7:7    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop8         7:8    0 174.6M  1 loop /snap/mesa-2404/1166
loop9         7:9    0  17.5M  1 loop /snap/prompting-client/221
loop10        7:10   0 552.9M  1 loop /snap/gnome-46-2404/154
loop11        7:11   0  17.5M  1 loop /snap/prompting-client/203
loop12        7:12   0  14.9M  1 loop /snap/snap-store/1368
loop13        7:13   0  10.9M  1 loop /snap/snap-store/1391
loop14        7:14   0  42.6M  1 loop /snap/snapd/26869
loop15        7:15   0   560K  1 loop /snap/snapd-desktop-integration/363
nvme0n1     259:0    0    30G  0 disk
├─nvme0n1p1 259:1    0     1G  0 part /boot/efi
└─nvme0n1p2 259:2    0  28.9G  0 part /
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ lsblk -f

NAME        FSTYPE   FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       squashfs 4.0                                                    0   100% /snap/core24/1588
loop1       squashfs 4.0                                                    0   100% /snap/bare/5
loop2       squashfs 4.0                                                    0   100% /snap/core24/1644
loop3       squashfs 4.0                                                    0   100% /snap/desktop-security-center/149
loop4       squashfs 4.0                                                    0   100% /snap/desktop-security-center/152
loop5       squashfs 4.0                                                    0   100% /snap/firefox/8105
loop6       squashfs 4.0                                                    0   100% /snap/firefox/8801
loop7       squashfs 4.0                                                    0   100% /snap/gtk-common-themes/1535
loop8       squashfs 4.0                                                    0   100% /snap/mesa-2404/1166
loop9       squashfs 4.0                                                    0   100% /snap/prompting-client/221
loop10      squashfs 4.0                                                    0   100% /snap/gnome-46-2404/154
loop11      squashfs 4.0                                                    0   100% /snap/prompting-client/203
loop12      squashfs 4.0                                                    0   100% /snap/snap-store/1368
loop13      squashfs 4.0                                                    0   100% /snap/snap-store/1391
loop14      squashfs 4.0                                                    0   100% /snap/snapd/26869
loop15      squashfs 4.0                                                    0   100% /snap/snapd-desktop-integration/363
nvme0n1
├─nvme0n1p1 vfat     FAT32       0020-912D                                 1G     1% /boot/efi
└─nvme0n1p2 ext4     1.0         3f976e8c-039d-439e-af77-9a6ff16d55db   16.2G    37% /
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ df -T
Filesystem     Type     1K-blocks     Used Available Use% Mounted on
tmpfs          tmpfs       691904     2256    689648   1% /run
/dev/nvme0n1p2 ext4      29701092 11135892  17031096  40% /
tmpfs          tmpfs      1729760        0   1729760   0% /dev/shm
```



```
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ df -T
Filesystem     Type     1K-blocks     Used Available Use% Mounted on
tmpfs          tmpfs       691904     2256    689648   1% /run
/dev/nvme0n1p2 ext4      29701092 11135892  17031096  40% /
tmpfs          tmpfs      1729760        0   1729760   0% /dev/shm
efivarfs       efivarfs       256       34       223  14% /sys/firmware/efi/efivars
none           tmpfs         1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs          tmpfs      1729760        8   1729752   1% /tmp
none           tmpfs         1024        0      1024   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p1 vfat       1098632     6680   1091952   1% /boot/efi
tmpfs          tmpfs       345952       76    345876   1% /run/user/1000
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           676M  2.2M  674M   1% /run
/dev/nvme0n1p2   29G   11G   17G  40% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
efivarfs        256K   34K  223K  14% /sys/firmware/efi/efivars
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.7G  8.0K  1.7G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p1  1.1G  6.6M  1.1G   1% /boot/efi
tmpfs           338M   76K  338M   1% /run/user/1000
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p2   29G   11G   17G  40% /
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$
```

| **Device/Partition** | **Filesystem Type** | **Size** | **Mount Point** |
|---|---|---|---|
| /dev/sda1 | Ext4 | 19G | / |
| /dev/sda2 | swap | 2G | [swap] |
| tmpfs | tmfs | ~1.6G | /run |
| tmpfs | tmpfs | ~7.7G | /dev/shm |
| tmpfs | tmpfs | ~5.0M | /run/lock |
| tmpfs | tmpfs | ~1.6G | /run/user/1000 |

## Exercise 2 – Disk → Partition → Filesystem → Mount Point

**Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.**

Linux Storage Organization

/dev/sda

|— /dev/sda1   ext4   ~19G   mounted at /

|— /dev/sda2   swap   ~2G    used as swap

The physical disk sda is divided into separate partitions such as sda1, sda2, The sda1 partition uses the ext4 filesystem and is mounted at /, while sda2 uses the swap filesystem for swap space.

### Exercise 3 – Explore the Root Filesystem

**Start from your current location.**

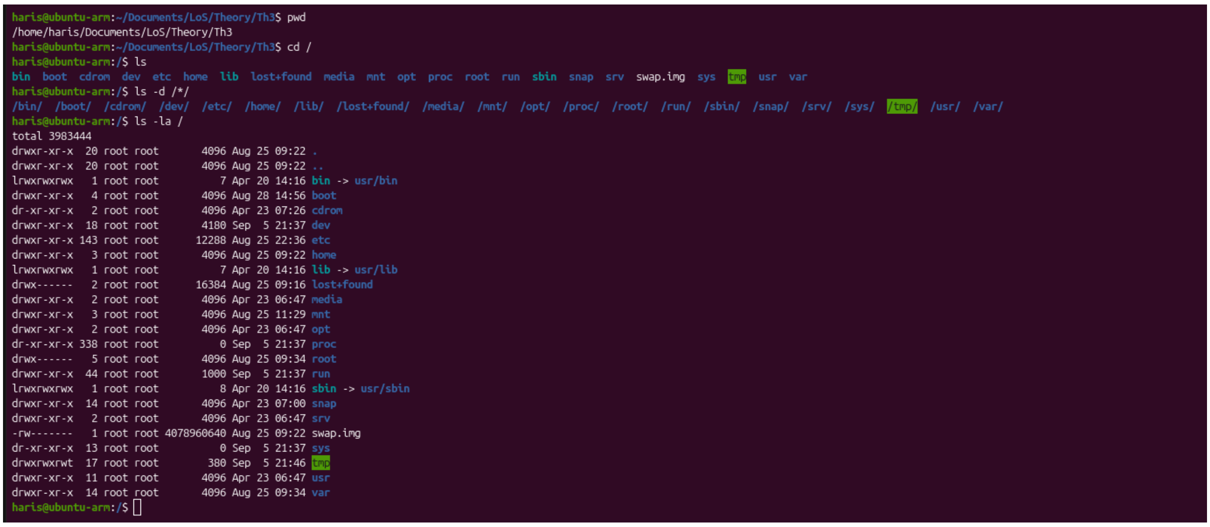
**1. Determine your current working directory.**

**2. Navigate to the root directory.**

**3. Display the contents of the root directory.**

**4. Identify all directories directly available under `/`.**

**5. Determine whether hidden entries are present.**



```
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ pwd
/home/haris/Documents/LoS/Theory/Th3
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ cd /
haris@ubuntu-arm:/$ ls
bin  boot  cdrom  dev  etc  home  lib  lost+found  media  mnt  opt  proc  root  run  sbin  snap  srv  swap.img  sys  tmp  usr  var
haris@ubuntu-arm:/$ ls -d /*/
/bin/  /boot/  /cdrom/  /dev/  /etc/  /home/  /lib/  /lost+found/  /media/  /mnt/  /opt/  /proc/  /root/  /run/  /sbin/  /snap/  /srv/  /sys/  /tmp/  /usr/  /var/
haris@ubuntu-arm:/$ ls -la /
total 3983444
drwxr-xr-x  20 root root       4096 Aug 25 09:22 .
drwxr-xr-x  20 root root       4096 Aug 25 09:22 ..
lrwxrwxrwx   1 root root          7 Apr 20 14:16 bin -> usr/bin
drwxr-xr-x   4 root root       4096 Aug 28 14:56 boot
dr-xr-xr-x   2 root root       4096 Apr 23 07:26 cdrom
drwxr-xr-x  18 root root       4180 Sep  5 21:37 dev
drwxr-xr-x 143 root root      12288 Aug 25 22:36 etc
drwxr-xr-x   3 root root       4096 Aug 25 09:22 home
lrwxrwxrwx   1 root root          7 Apr 20 14:16 lib -> usr/lib
drwx------   2 root root      16384 Aug 25 09:16 lost+found
drwxr-xr-x   2 root root       4096 Apr 23 06:47 media
drwxr-xr-x   3 root root       4096 Aug 25 11:29 mnt
drwxr-xr-x   2 root root       4096 Apr 23 06:47 opt
dr-xr-xr-x 338 root root          0 Sep  5 21:37 proc
drwx------   5 root root       4096 Aug 25 09:34 root
drwxr-xr-x  44 root root       1000 Sep  5 21:37 run
lrwxrwxrwx   1 root root          8 Apr 20 14:16 sbin -> usr/sbin
drwxr-xr-x  14 root root       4096 Apr 23 07:00 snap
drwxr-xr-x   2 root root       4096 Apr 23 06:47 srv
-rw-------   1 root root 4078960640 Aug 25 09:22 swap.img
dr-xr-xr-x  13 root root          0 Sep  5 21:37 sys
drwxrwxrwt  17 root root        380 Sep  5 21:46 tmp
drwxr-xr-x  11 root root       4096 Apr 23 06:47 usr
drwxr-xr-x  14 root root       4096 Aug 25 09:34 var
haris@ubuntu-arm:/$
```

### Exercise 4 – Linux Filesystem Hierarchy Investigation

**For each location:**

**1. Examine its contents.**

**2. Identify at least three entries, where available.**

**3. Determine the likely purpose of the directory.**

| **Directory** | **Example Contents** | **Purpose** |
|---|---|---|
| /home | User directories, ., .. | Stores the personal folders of regular users.. |
| /root | .bashrc, .zshrc, .ssh | Stores the superuser's home directory and personal settings. |
| /etc | passwd, shadow, hosts, fstab | Holds system-wide configuration files for the OS and services. |
| /var | log, cache, spool | Contains data that changes during normal system use, such as logs, caches, and spool files. |
| /var/log | boot.log, wtmp, btmp, journal/ | Stores logs for the system, boot process, logins, and applications. |
| /tmp | .X11-unix,ICE-unix, temporary directories | Provides short-term storage for temporary files and session-related data. |
| /usr | bin, lib, share | Contains installed applications, libraries, and shared documentation or data. |
| /boot | Kernel files, initrd, grub/ | Holds the kernel, initial RAM filesystem, and bootloader files needed to start the system |

| | | |
|---|---|---|
| /dev | null, zero, tty | Represents hardware and virtual devices as files. |
| /proc | cpuinfo, meminfo, self, PID directories | A virtual filesystem that provides live process and system information. |

**Exercise 5 – Find Your Home Directory**

**Without assuming its location:**

**1. Determine your username.**

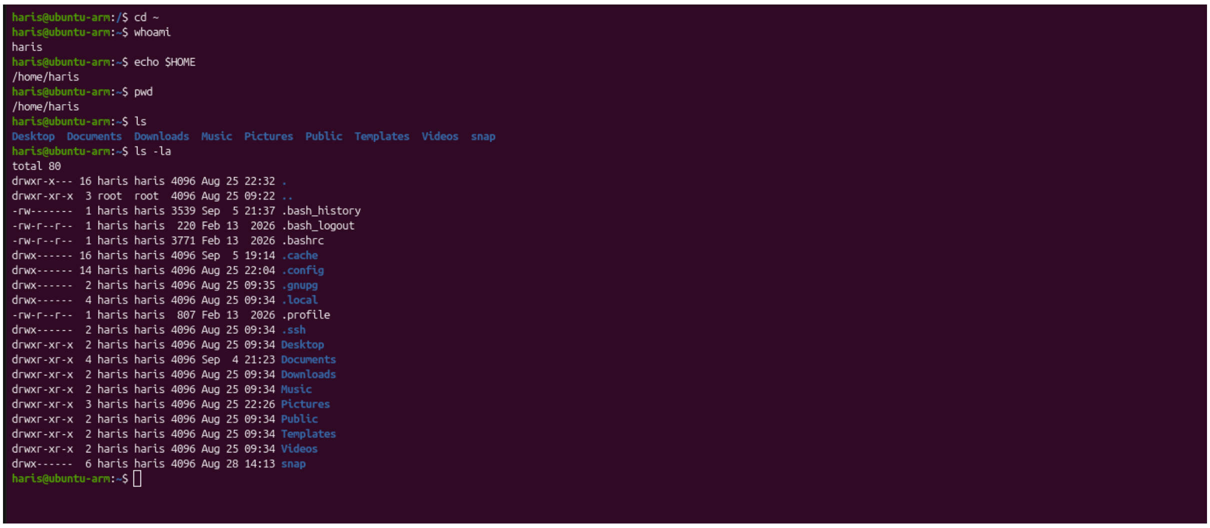
**2. Determine your home directory.**

**3. Navigate to your home directory.**

**4. Display its absolute path.**

**5. List its normal contents.**

**6. List all contents, including hidden files.**



```
haris@ubuntu-arm:/$ cd ~
haris@ubuntu-arm:~$ whoami
haris
haris@ubuntu-arm:~$ echo $HOME
/home/haris
haris@ubuntu-arm:~$ pwd
/home/haris
haris@ubuntu-arm:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
haris@ubuntu-arm:~$ ls -la
total 80
drwxr-x--- 16 haris haris 4096 Aug 25 22:32 .
drwxr-xr-x  3 root  root  4096 Aug 25 09:22 ..
-rw-------  1 haris haris 3539 Sep  5 21:37 .bash_history
-rw-r--r--  1 haris haris  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 haris haris 3771 Feb 13  2026 .bashrc
drwx------ 16 haris haris 4096 Sep  5 19:14 .cache
drwx------ 14 haris haris 4096 Aug 25 22:04 .config
drwx------  2 haris haris 4096 Aug 25 09:35 .gnupg
drwx------  4 haris haris 4096 Aug 25 09:34 .local
-rw-r--r--  1 haris haris  807 Feb 13  2026 .profile
drwx------  2 haris haris 4096 Aug 25 09:34 .ssh
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Desktop
drwxr-xr-x  4 haris haris 4096 Sep  4 21:23 Documents
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Downloads
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Music
drwxr-xr-x  3 haris haris 4096 Aug 25 22:26 Pictures
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Public
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Templates
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Videos
drwx------  6 haris haris 4096 Aug 28 14:13 snap
haris@ubuntu-arm:~$
```

**Exercise 6 – Hidden Files Investigation**

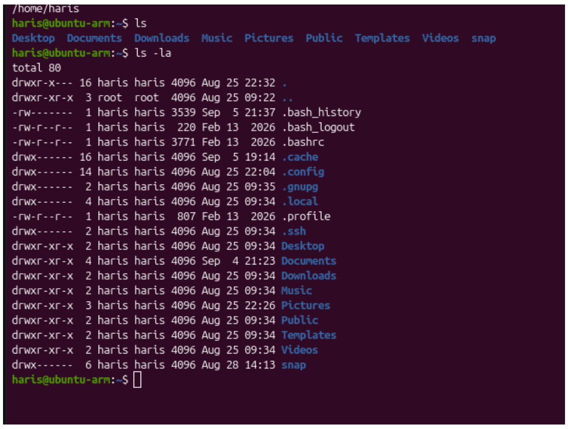
**Examine your home directory.**

**Compare:**

**ls ls -**

**a**

**Identify at least 2 hidden files/directories.**



```
/home/haris
haris@ubuntu-arm:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
haris@ubuntu-arm:~$ ls -la
total 80
drwxr-x--- 16 haris haris 4096 Aug 25 22:32 .
drwxr-xr-x  3 root  root  4096 Aug 25 09:22 ..
-rw-------  1 haris haris 3539 Sep  5 21:37 .bash_history
-rw-r--r--  1 haris haris  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 haris haris 3771 Feb 13  2026 .bashrc
drwx------ 16 haris haris 4096 Sep  5 19:14 .cache
drwx------ 14 haris haris 4096 Aug 25 22:04 .config
drwx------  2 haris haris 4096 Aug 25 09:35 .gnupg
drwx------  4 haris haris 4096 Aug 25 09:34 .local
-rw-r--r--  1 haris haris  807 Feb 13  2026 .profile
drwx------  2 haris haris 4096 Aug 25 09:34 .ssh
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Desktop
drwxr-xr-x  4 haris haris 4096 Sep  4 21:23 Documents
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Downloads
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Music
drwxr-xr-x  3 haris haris 4096 Aug 25 22:26 Pictures
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Public
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Templates
drwxr-xr-x  2 haris haris 4096 Aug 25 09:34 Videos
drwx------  6 haris haris 4096 Aug 28 14:13 snap
haris@ubuntu-arm:~$
```

**Answer:**

1. **How does Linux identify a hidden file?**

**Any file or directory whose name starts with a dot (.) is treated as hidden in Linux.**

2. **Why does a normal `ls` not display these files?**

**The normal ls command does not show files that start with a dot by default. This is simply to keep the usual listing clean and avoid showing configuration and other hidden files. It is only a display convention, not a security feature.**

3. **What do `.` and `..` mean?**

**`.` represents the current directory, while `..` represents the parent directory.**

4. **Can a hidden file still be accessed if its name is known?**

**Yes. A hidden file can still be accessed normally if you know its name. The dot at the beginning only makes it hidden from the usual ls listing; it does not stop you from opening or using the file.**

5. **From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?**

**Hidden files are not always safe or harmful just because they are hidden. Some, like .bashrc and .bash_history, are normal system files, while .ssh can contain sensitive information. So, an investigator should check the file's contents, permissions, and purpose before deciding whether it is suspicious.**

## Exercise 7 – Build a Small Filesystem Hierarchy

**Explore: mkdir**

**Hint: you can create this hierarchy in a single command**

**Create the following files:**

**cyber_lab/evidence/network/traffic.txt**

**cyber_lab/evidence/system/processes.txt**

**cyber_lab/logs/security.log cyber_lab/scripts/check.sh**

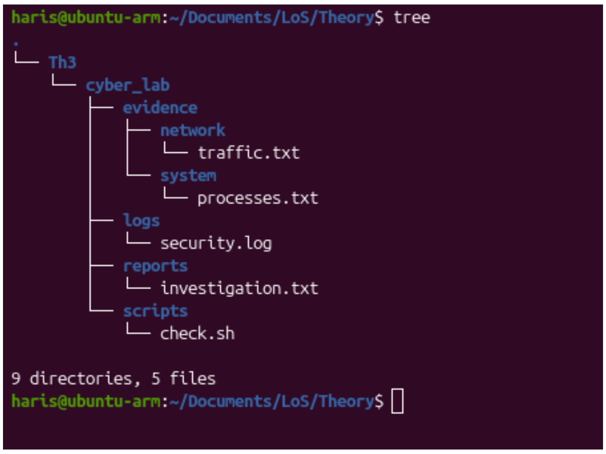
**cyber_lab/reports/investigation.txt**

**Use appropriate Linux commands to create the structure.**

**Verify that the required hierarchy has been created successfully.**

**Hint: tree**

**Explore options**




**Exercise 8 – Absolute and Relative Paths**

**Assume your current directory is:**

**~/cyber_lab**

**Perform the following:**

**1. Navigate to `evidence/network` using a relative path.**

**2. Navigate to `/etc` using an absolute path.**

**3. Return to your home directory using the shortest possible command.**

**4. Navigate to `cyber_lab/reports`.**

**5. Move one directory upward.**

**6. Display the current directory after every operation.**

```
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ pwd
/home/haris/Documents/LoS/Theory/Th3
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3$ cd cyber_lab/
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3/cyber_lab$ cd ./evidence/
network/ system/
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3/cyber_lab$ cd ./evidence/network
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3/cyber_lab/evidence/network$ cd /etc
haris@ubuntu-arm:/etc$ cd ~/
.cache/    .config/   Desktop/   Documents/ Downloads/ .gnupg/    .local/    Music/     Pictures/  Public/    snap/      .ssh/      Templates/ Videos/
haris@ubuntu-arm:/etc$ cd ~
haris@ubuntu-arm:~$ cd Documents/LoS/Theory/Th3/cyber_lab/reports/
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3/cyber_lab/reports$ cd ..
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3/cyber_lab$ pwd
/home/haris/Documents/LoS/Theory/Th3/cyber_lab
haris@ubuntu-arm:~/Documents/LoS/Theory/Th3/cyber_lab$
```

*** What is an absolute path?**

An absolute path starts from the root directory / and gives the complete location of a file or directory. For example:

/home/haris/Documents/LoS/Theory/Th3/cyber_lab/evidence/network

*** What is a relative path?**

A relative path specifies a location based on the current working directory. For example:

evidence/network.

*** What is the difference between `/` and `~`?**

/ : root directory of the entire Linux filesystem.

~ : shortcut for the current user's home directory.

**Exercise 9 – Identify File Types**

**Investigate the following:**

**/etc/passwd**

**/etc**

**/bin/bash**

**/dev/null**

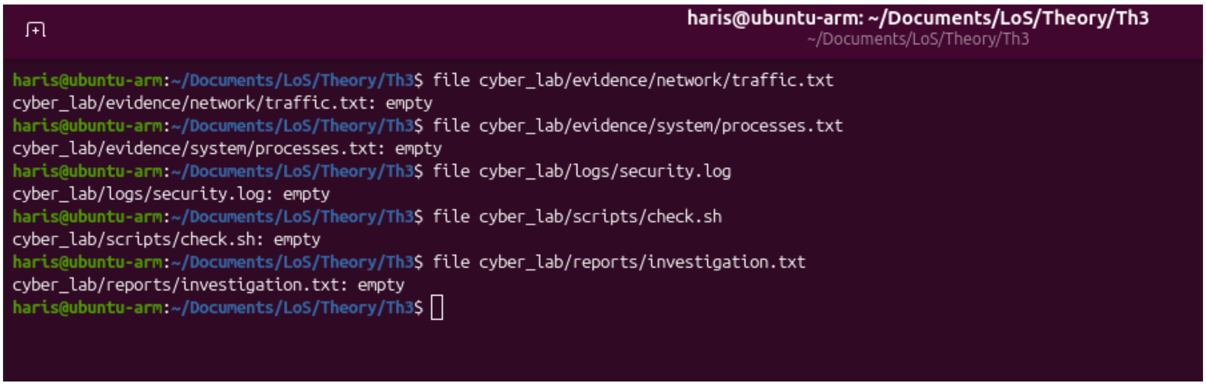
**/proc**

**Also investigate the files created in Exercise 7.**

**file <filename> ls -**

**l <filename>**

**Classify each object.**


| **Object** | **Type** |
|---|---|
| /etc/passwd | Regular text file |
| /etc | Directory |
| /bin/bash | Executable files |
| /dev/null | Character special device file |
| /proc | Virtual/pseudo filesystem directory |
| traffic.txt | Regular files |
| processes.txt | Regular files |
| security.log | Regular files |
| check.sh | Regular files |
| investigation.txt | Regular files |

**Exercise 10 – Examine File Metadata**

**Use:~/cyber_lab/reports/investigation.txt**

**Add a small amount of text to the file.**

**Examine it using:**

**ls -l**

**stat investigation.txt**

**Identify:**

*** Filename**

*** File size**

*** File type**

*** Owner**

*** Group**

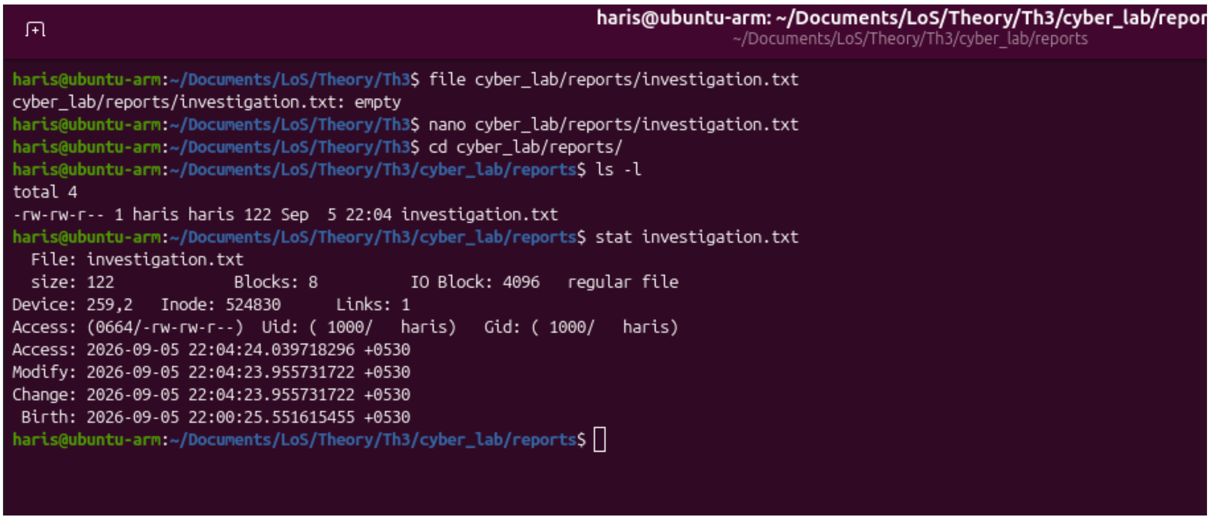
*** Inode number**

*** Access time**

*** Modification time**

*** Change time**

**Answer:**




1. **What is metadata?**

Metadata is information that describes a file, such as its size, owner, permissions, inode number, and time stamps.

2. **Is metadata the same as file content?**

No. Metadata describes the file, while file content is the actual data stored inside the file.

### 3. Which timestamp changes when file contents are modified?

The modification time, or mtime, changes when a file's contents are edited.

**Exercise 11 – Explore `/etc`**

**Investigate:**

**/etc**

` ` `

**Locate:**

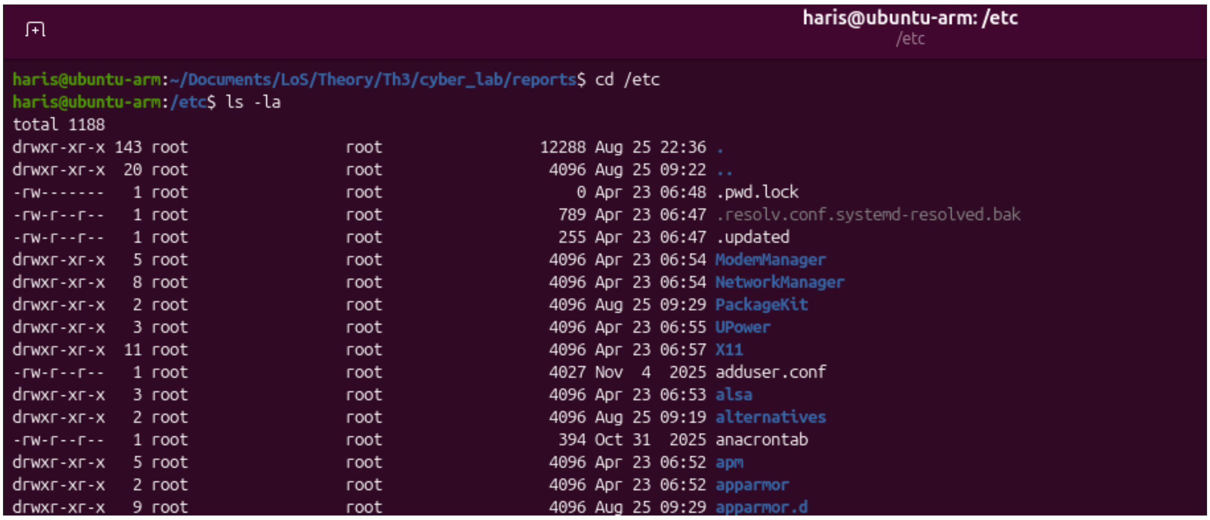
**passwd group**

**hostname**

**hosts**

**Note: Without modifying any system files, examine their contents where permitted.**

**Answer:**



```
haris@ubuntu-arm: /etc
/etc

haris@ubuntu-arm:~/Documents/LoS/Theory/Th3/cyber_lab/reports$ cd /etc
haris@ubuntu-arm:/etc$ ls -la
total 1188
drwxr-xr-x 143 root                root                12288 Aug 25 22:36 .
drwxr-xr-x  20 root                root                 4096 Aug 25 09:22 ..
-rw-------   1 root                root                    0 Apr 23 06:48 .pwd.lock
-rw-r--r--   1 root                root                  789 Apr 23 06:47 .resolv.conf.systemd-resolved.bak
-rw-r--r--   1 root                root                  255 Apr 23 06:47 .updated
drwxr-xr-x   5 root                root                 4096 Apr 23 06:54 ModemManager
drwxr-xr-x   8 root                root                 4096 Apr 23 06:54 NetworkManager
drwxr-xr-x   2 root                root                 4096 Aug 25 09:29 PackageKit
drwxr-xr-x   3 root                root                 4096 Apr 23 06:55 UPower
drwxr-xr-x  11 root                root                 4096 Apr 23 06:57 X11
-rw-r--r--   1 root                root                 4027 Nov  4  2025 adduser.conf
drwxr-xr-x   3 root                root                 4096 Apr 23 06:53 alsa
drwxr-xr-x   2 root                root                 4096 Aug 25 09:19 alternatives
-rw-r--r--   1 root                root                  394 Oct 31  2025 anacrontab
drwxr-xr-x   5 root                root                 4096 Apr 23 06:52 apm
drwxr-xr-x   2 root                root                 4096 Apr 23 06:52 apparmor
drwxr-xr-x   9 root                root                 4096 Aug 25 09:29 apparmor.d
```

```
haris@ubuntu-arm: /etc
/etc
haris@ubuntu-arm:/etc$ cat hostname
ubuntu-arm
haris@ubuntu-arm:/etc$ cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
apt:x:42:65534::/nonexistent:/usr/sbin/nologin
```

```
haris@ubuntu-arm: /etc
/etc
gdm:x:975:975:Gnome Display Manager:/var/lib/gdm3:/bin/false
haris:x:1000:1000:Haris:/home/haris:/bin/bash
haris@ubuntu-arm:/etc$ cat hosts
127.0.0.1 localhost
127.0.1.1 ubuntu-arm

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
haris@ubuntu-arm:/etc$ cat group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,haris
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:haris
floppy:x:25:
tape:x:26:
sudo:x:27:haris
audio:x:29:
dip:x:30:haris
www-data:x:33:
backup:x:34:
operator:x:37:
list:x:38:
irc:x:39:
src:x:40:
shadow:x:42:
utmp:x:43:
```

**1. What kind of information is primarily stored under `/etc`?**

/etc mainly stores system settings and configuration files.

**2. What information does `/etc/passwd` contain?**

/etc/passwd contains information about the users on the system.

**3. What information does `/etc/hostname` contain?**

/etc/hostname contains the name of the computer/system.

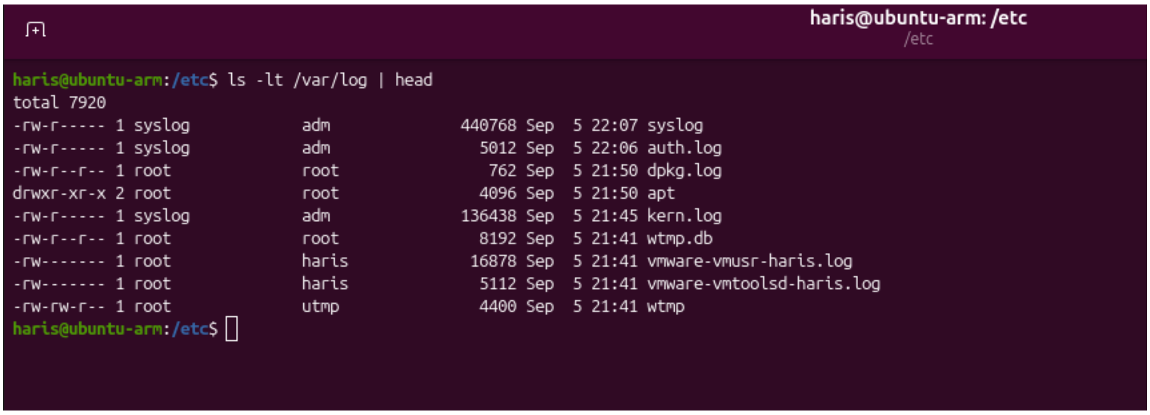
**4. Why is `/etc` security-sensitive?**

/etc is security-sensitive because it contains important system and user settings.

**Exercise 12 – Explore Linux Log Storage**

**Investigate:**

**/var/log**

```

**Identify:**

**\* Available log files**

**\* Available log directories**

**\* Recently modified entries**

**\* Entries you can read**

**\* Entries you cannot read as a normal user**

**Answer:**




Identify

**Available log files:** syslog, auth.log, boot.log, kern.log, wtmp, dpkg.log, `apt

**Available log directories:** apt/ and journal/ (if present in /var/log)

**Recently modified entries:** syslog, auth.log, boot.log, kern.log, wtmp, vmware-vmtoolsd-haris.log, vmware-vmtoolsd-root.log
```

**Entries you can read as a normal user:** files with read permission for your user, such as syslog, auth.log, boot.log, kern.log, wtmp, and the VMware log files shown with -rw-r-----

**Entries you cannot read as a normal user:** root-owned logs with restricted permissions, especially files showing -rw------- or -rw-r----- without group access

**1. Why does Linux maintain logs?**

Linux maintains logs to record system and user activities, errors, and events.

**2. Why are logs useful during incident investigation?**

Logs help investigators find out what happened and when it happened.

**3. Why might access to some logs be restricted?**

Some logs may contain sensitive user or security information, so access is restricted to protect them.

**Exercise 13 – Investigate `/tmp`**

**Explore `/tmp` without modifying files belonging to other users.**

**Determine:**

**1. Whether files/directories currently exist.**

Yes. /tmp has several current files and folders used for temporary data, sockets, and session-related information.

**2. Whether hidden files exist.**

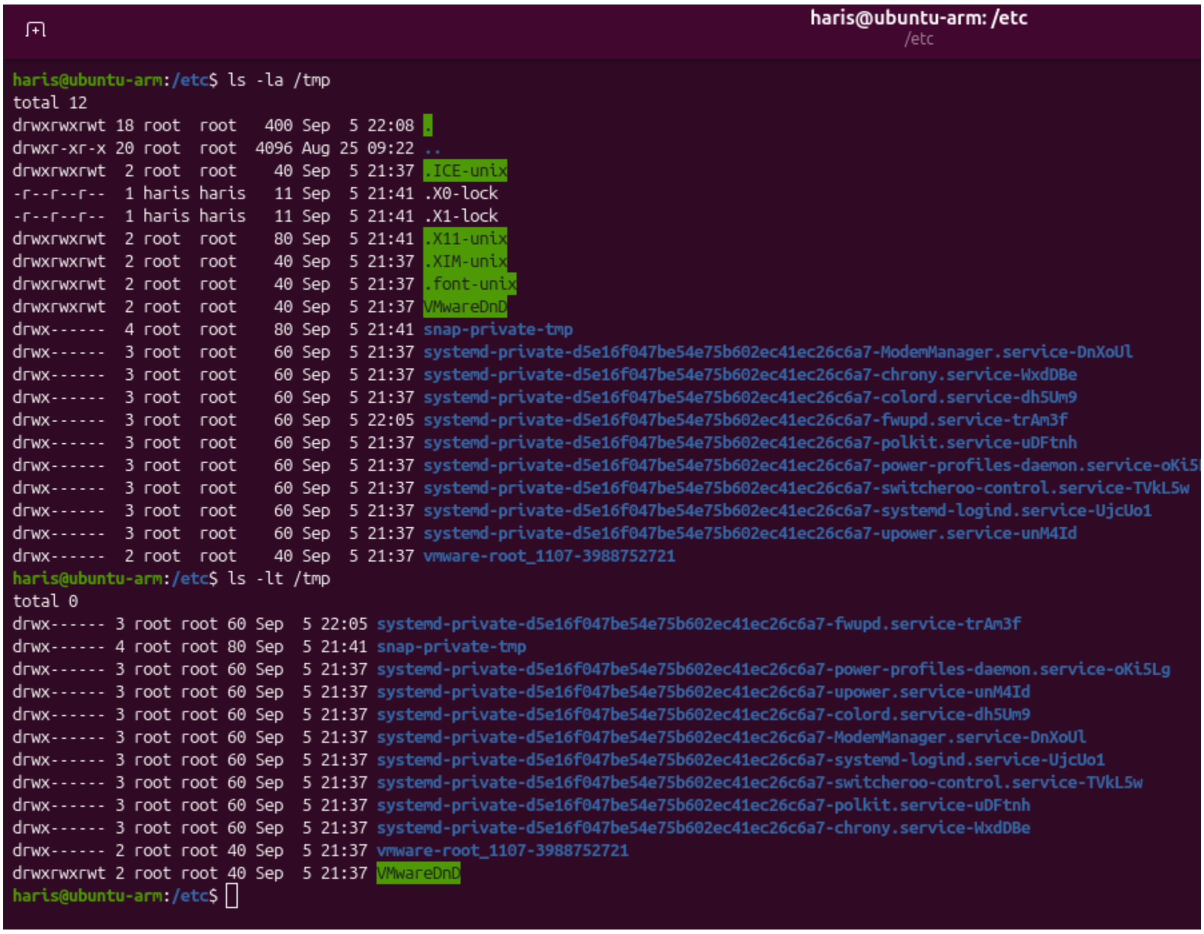
Yes. It can include hidden items such as .X11-unix, .ICE-unix, and other dot-named entries.

**3. Who owns some of the entries.**

Ownership varies by entry; some may belong to root and others to the user or process that created them.

**4. Which entries were modified recently.**

ls -lt /tmp lists items by last modification time, with the newest entries at the top.



```
haris@ubuntu-arm: /etc
/etc
haris@ubuntu-arm:/etc$ ls -la /tmp
total 12
drwxrwxrwt 18 root  root   400 Sep  5 22:08 .
drwxr-xr-x 20 root  root  4096 Aug 25 09:22 ..
drwxrwxrwt  2 root  root    40 Sep  5 21:37 .ICE-unix
-r--r--r--  1 haris haris   11 Sep  5 21:41 .X0-lock
-r--r--r--  1 haris haris   11 Sep  5 21:41 .X1-lock
drwxrwxrwt  2 root  root    80 Sep  5 21:41 .X11-unix
drwxrwxrwt  2 root  root    40 Sep  5 21:37 .XIM-unix
drwxrwxrwt  2 root  root    40 Sep  5 21:37 .font-unix
drwxrwxrwt  2 root  root    40 Sep  5 21:37 VMwareDnD
drwx------  4 root  root    80 Sep  5 21:41 snap-private-tmp
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-ModemManager.service-DnXoUl
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-chrony.service-WxdDBe
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-colord.service-dh5Um9
drwx------  3 root  root    60 Sep  5 22:05 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-fwupd.service-trAm3f
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-polkit.service-uDFtnh
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-power-profiles-daemon.service-oKi5
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-switcheroo-control.service-TVkL5w
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-systemd-logind.service-UjcUo1
drwx------  3 root  root    60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-upower.service-unM4Id
drwx------  2 root  root    40 Sep  5 21:37 vmware-root_1107-3988752721
haris@ubuntu-arm:/etc$ ls -lt /tmp
total 0
drwx------ 3 root root 60 Sep  5 22:05 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-fwupd.service-trAm3f
drwx------ 4 root root 80 Sep  5 21:41 snap-private-tmp
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-power-profiles-daemon.service-oKi5Lg
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-upower.service-unM4Id
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-colord.service-dh5Um9
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-ModemManager.service-DnXoUl
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-systemd-logind.service-UjcUo1
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-switcheroo-control.service-TVkL5w
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-polkit.service-uDFtnh
drwx------ 3 root root 60 Sep  5 21:37 systemd-private-d5e16f047be54e75b602ec41ec26c6a7-chrony.service-WxdDBe
drwx------ 2 root root 40 Sep  5 21:37 vmware-root_1107-3988752721
drwxrwxrwt 2 root root 40 Sep  5 21:37 VMwareDnD
haris@ubuntu-arm:/etc$
```

**Answer:**

**Why can temporary directories be relevant during a cybersecurity investigation?**

`/tmp` is important because many programs use it to store temporary files. In a security investigation, it may contain suspicious scripts, downloaded items, extracted archives, or malware-related files, so it should be checked closely.

**Exercise 14 – Explore the `/proc` Filesystem**

**Investigate:**

**/proc**

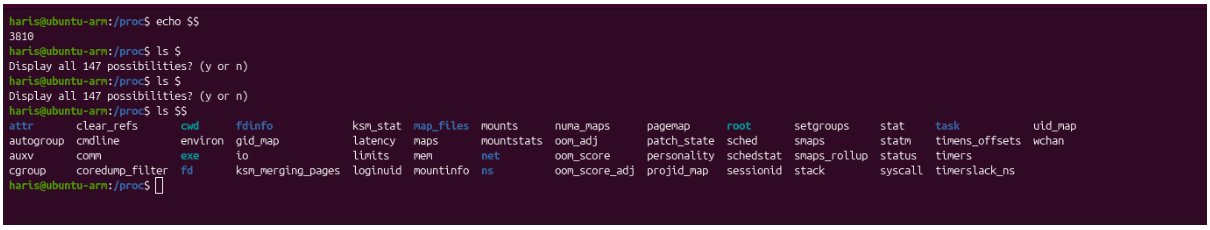
```

**Examine:**

**cat /proc/cpuinfo cat**

**/proc/meminfo ```**
```

**Then identify your current shell PID: echo**

**$$**

```
```

**Check whether a corresponding directory exists under `/proc`.**


**Answer:**

**1. What information does `/proc/cpuinfo` provide?**

/proc/cpuinfo shows details about the CPU, including its model, clock speed, cache, and supported capabilities.

**2. What information does `/proc/meminfo` provide?**

/proc/meminfo gives information about the system's memory usage, including RAM and swap.

**3. Why do directories with numbers appear inside `/proc`?**

The numbered folders in /proc are process IDs for running programs.

**4. Is `/proc` an ordinary disk directory?**

No. /proc is a virtual filesystem created by the kernel, not a normal disk-based directory.

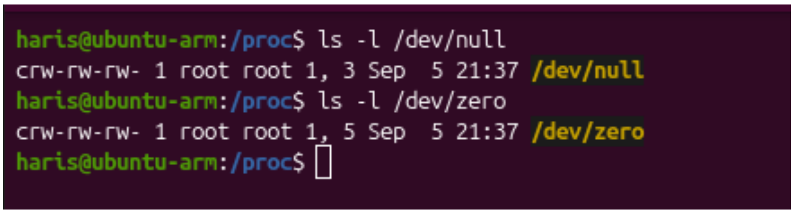
**5. Why could `/proc` be useful during security investigation?**

/proc is useful in investigations because it reveals active processes and system behavior, which can help spot suspicious activity.

**Exercise 15 – Explore `/dev`**

**/dev**

```
Examine:

ls -l /dev/null ls -

l /dev/zero
```




**Answer:**

**1. What does `/dev` represent?**

It is where Linux provides hardware and virtual devices as files. Programs can interact with devices such as disks and terminals through these files.

**2. Are the entries ordinary user files?**

No. They are character devices, not regular files. The c at the beginning of the permissions shows that they are character devices.

**3. Determine the file type of `/dev/null`.**

/dev/null is a character special device with major number 1 and minor number 3.

**4. Why does Linux expose devices through the filesystem?**

Linux treats devices much like files, which makes it easy for programs to interact with them. For example, /dev/null simply throws away anything sent to it, while /dev/zero keeps giving zeros whenever it is read.

**Exercise 16 – Filesystem Space Investigation**

**Determine:**

**1. Total filesystem capacity**

29G

**2. Used space.**

11G

**3. Available space.**

17G

**4. Percentage of filesystem utilization.**

40%

**5. Space occupied by your home directory.**

185M

**6. Space occupied by the `cyber_lab` directory.**

32K

**Use appropriate combinations of:**

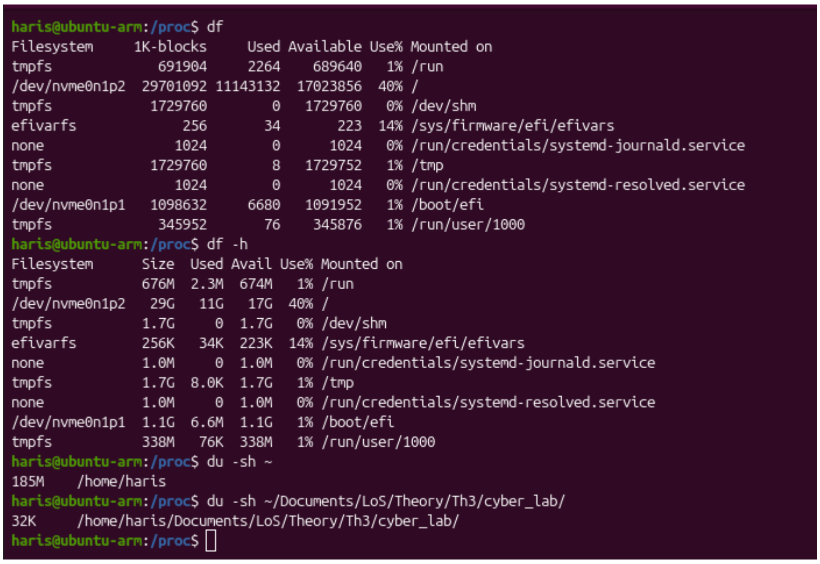
**df**

**df h**

**du**

**du -h**

**du -sh**



```
haris@ubuntu-arm:/proc$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
tmpfs             691904     2264    689640   1% /run
/dev/nvme0n1p2  29701092 11143132  17023856  40% /
tmpfs            1729760        0   1729760   0% /dev/shm
efivarfs             256       34       223  14% /sys/firmware/efi/efivars
none                1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            1729760        8   1729752   1% /tmp
none                1024        0      1024   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p1   1098632     6680   1091952   1% /boot/efi
tmpfs             345952       76    345876   1% /run/user/1000
haris@ubuntu-arm:/proc$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           676M  2.3M  674M   1% /run
/dev/nvme0n1p2   29G   11G   17G  40% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
efivarfs        256K   34K  223K  14% /sys/firmware/efi/efivars
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.7G  8.0K  1.7G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p1  1.1G  6.6M  1.1G   1% /boot/efi
tmpfs           338M   76K  338M   1% /run/user/1000
haris@ubuntu-arm:/proc$ du -sh ~
185M    /home/haris
haris@ubuntu-arm:/proc$ du -sh ~/Documents/LoS/Theory/Th3/cyber_lab/
32K     /home/haris/Documents/LoS/Theory/Th3/cyber_lab/
haris@ubuntu-arm:/proc$ 
```

**Answer:**

**Explain the difference between `df` and `du`.**

*`df` shows disk space usage for an entire filesystem. `du` shows how much space a specific file or directory uses.*

**Exercise 17 – Cybersecurity Investigation Scenario**

**Assume that you are investigating a Linux machine after suspicious activity has been reported.**

**You are initially asked to examine only the filesystem.**

**Identify which of the following locations you would investigate and explain **why**:**

**/home**

**/etc**

**/var/log**

**/tmp**

**/proc**

**/dev**

**/root**

**Answers:**

**/home**: useful for checking suspicious files, downloads, hidden items, scripts, and signs of user activity.

**/etc:** useful for spotting altered system settings, new accounts, sudo changes, or scheduled jobs.

**/var/log:** useful for reviewing logins, errors, and other traces of activity.

**/tmp:** may contain temporary malicious files or scripts used by an attacker.

**/proc:** shows currently running processes, which can reveal suspicious activity.

**/root:** if accessible, it may indicate that root access was obtained.